# Jane Doe

(555) 555-0142 | jane.doe@example.com | linkedin.com/in/jane-doe-example | github.com/jane-doe-example
Springfield, ST | US Citizen | Willing to relocate

## Education

**B.S. Electrical Engineering**, GPA 3.62/4.00 Expected Spring 2027

Example State University

**Relevant coursework:** Power Electronics, Embedded Systems, Digital Design, Signals and Systems

**Activities:** IEEE student branch, Tau Beta Pi

## Projects

**Synchronous Buck Converter for a CubeSat Power Board** Fall 2025–Spring 2026

*KiCad, LTspice, STM32, C*

- Reached 92% peak efficiency at a 2 A load, measured on an electronic load, by replacing the catch diode with a synchronous MOSFET stage and tuning dead time with an STM32 timer
- Cut output ripple from 80 mV to 25 mV peak to peak, measured on an oscilloscope, by redesigning the 4-layer PCB in KiCad with a solid ground plane and a tighter switching loop
- Wrote STM32 firmware in C that shuts the rail down within 50 µs of an overcurrent fault, verified over 20 deliberate short-circuit tests

**FPGA Audio Equalizer** Spring 2025

*Verilog, Vivado, MATLAB, Basys 3*

- Processed 48 kHz stereo audio at under 2 ms latency by pipelining five 64-tap FIR filters in Verilog
- Matched the MATLAB reference response within 0.5 dB from 20 Hz to 20 kHz by testing fixed-point coefficients in Vivado simulation

## Experience

**Power Subsystem Lead, Example State CubeSat Team**, Springfield, ST Sep 2024–Present

- Cut power board mass by 40 g by replacing three linear regulators with one buck converter board
- Found two battery protection faults before launch review by writing a Python script that runs overnight charge and discharge cycles and flags cell drift

Sales Associate, Example Hardware Store May 2023–Aug 2024

## Skills

**Software:** KiCad, LTspice, MATLAB, Vivado, Python, C, C++, Verilog

**Hardware/Lab:** Oscilloscope, Electronic load, STM32, ESP32, FPGA